

Lecture 24: Doping

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1 Introduction

Doping refers to the addition of specific *impurities* to a semiconductor to modify its electrical properties. In microfabrication the commonly used material is silicon. Pure Si (intrinsic Si) is a poor conductor with a negative temperature coefficient of resistance (i.e. resistance decreases with rise in temperature) due to thermal generation of electrons and holes. Doping helps in exponentially increasing the conductivity and also produces a stable, *temperature independent*, resistance around room temperature. Doped or extrinsic semiconductors can be p or n type depending on the nature of the impurity atom.

The starting wafers in integrated circuit (IC) manufacturing are usually p or n type doped wafers. To form devices like transistors, diodes, or resistors, specific regions of the wafer must be doped with specific amounts and types of impurities. Some examples of doped devices are shown in figure 1. There are two main methods of doping.

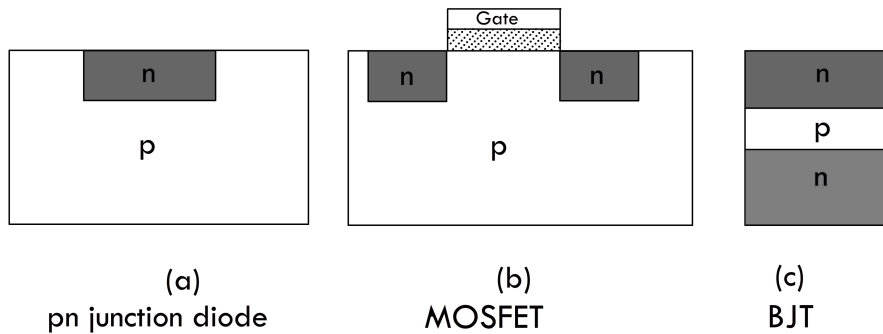


Figure 1: Schematic of a (a) diode, (b) MOSFET, and (c) BJT. These are made by adding p and n type dopants to different parts of a base wafer. In (a) and (b) the base wafer is p type while it is n type in (c).

1. Thermal diffusion
2. Ion implantation

The two methods are summarized in figure 2. As part of the process flow, there are specific goals that doping should meet

1. Create a specific concentration of dopant atoms at and below the surface of the wafer, i.e. establish a *controlled concentration gradient*.
2. Create a junction (pn or np or graded p or n) at a specific depth from the wafer surface. This is important for MOSFETs since this defines the channel width. In a BJT, doping is used to define widths of the emitter, base, and collector regions.
3. Create specific distributions and concentrations of dopants *laterally* along the wafer surface. This is related to patterning and is used to define the smallest lateral region that can be doped.

2 Thermal diffusion

Thermal diffusion is a two step process, similar to the steps in oxidation process by consuming the underlying Si.

1. **Deposition** - dopant atoms are introduced at the wafer surface.
2. **Drive-in** - the dopant atoms then diffuse into the wafer to create the required concentration gradient.

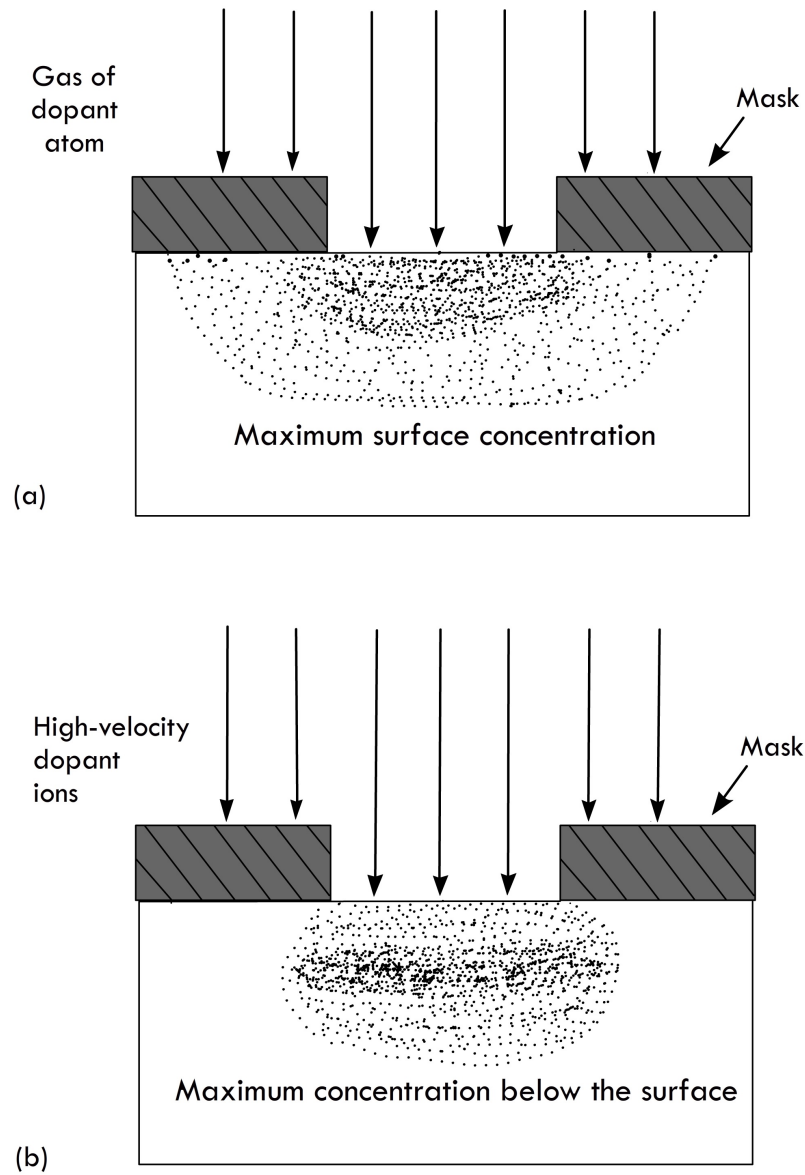


Figure 2: Types of doping. (a) Thermal diffusion (b) Ion implantation. In thermal diffusion, the maximum concentration is at the surface and dopants diffuse into the wafer. In ion implantation, the dopants are embedded below the surface. Adapted from *Microchip fabrication* - Peter van Zant.

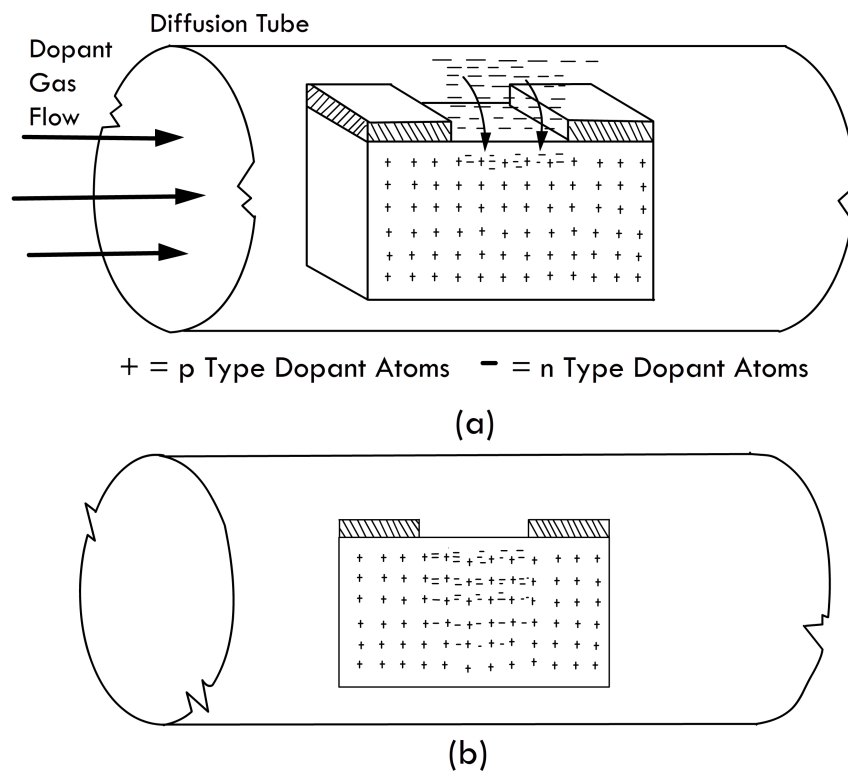


Figure 3: (a) The stages in thermal diffusion. The dopant atoms are introduced at the wafer surface and they diffuse into the wafer. (b) A cross section schematic showing the final concentration distribution. Adapted from *Microchip fabrication* - Peter van Zant.

The two steps are shown schematically in figure 3. While dopant atoms move vertically into the wafer, there is also a lateral spread. This has implications for the minimum dimensions of the region that can be doped. In Si, the common dopants are boron for p type and arsenic, antimony, and phosphorus for n type.

2.1 Diffusion sources

There are different sources for the dopant atoms. These can be solid, liquid, or gaseous sources. Some examples of dopant materials for Si are

1. Antimony (Sb) - Sb_2O_3 (s)
2. Arsenic (As) - As_2O_3 (s), AsH_3 (g)
3. Phosphorus (P) - POCl_3 (l), P_2O_5 (s), PH_3 (g)
4. Boron (B) - BBr_3 (l), B_2O_3 (s), BCl_3 (g)

A more complete list of sources are shown in table 1.

For liquid and gaseous sources, a concentration of the dopant vapor should

Table 1: Some commonly diffusion sources in silicon. The compound and its state is also included. The use of dopants in different state determines the shape of the concentration profile within the wafer. Adapted from *Microchip fabrication* - Peter van Zant.

Type	Element	Compound	Formula	State
n -type	Antimony Arsenic	Antimony Trioxide	Sb_2O_3	solid
		Arsenic Trioxide	As_2O_3	solid
		Arsine	AsH_3	gas
	Phosphorous	Phosphorous oxychloride	POCl_3	liquid
		Phosphorous Pentoxide	P_2O_5	solid
		Phosphine	PH_3	gas
p -type	Boron	Boron Tribromide	BBr_3	liquid
		Boron Trioxide	B_2O_3	solid
		Diborane	B_2H_6	gas
		Boron Trichloride	BCl_3	gas
		Boron Nitride	BN	solid

be established at the surface. For a liquid source, a carrier gas is usually used to transport the vapors to the diffusion furnace. The setup is shown

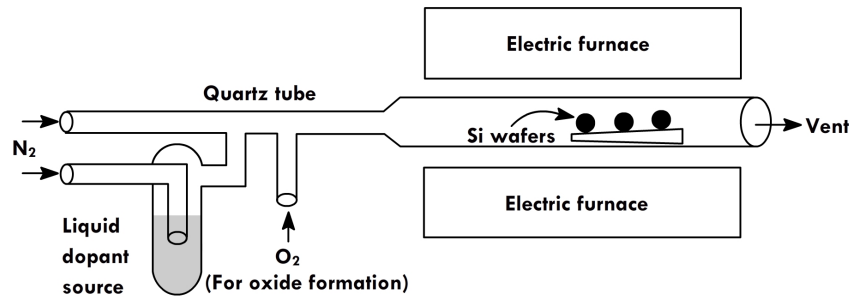


Figure 4: Thermal diffusion setup with a liquid dopant source. A carrier gas like nitrogen is bubbled through the dopant liquid and the vapor is carried into the furnace. Oxygen is also used when an oxide surface needs to be created along with doping. Adapted from *Fundamentals of semiconductor manufacturing and process control* - May and Spanos.

in figure 4. A similar arrangement is used for gaseous sources. Here, the carrier gas is used to dilute the dopant gas to the required concentration. The gas manifold system is shown in figure 5. For solid sources, wafer sized "slugs" are packed into the furnace along with the product wafers (e.g. for boron, boron nitride slugs can be used as solid sources), this is called a *solid neighbor source*. An other option is to spin on the oxide source on the wafer surface using a suitable solvent, typically used for oxide sources. It is also possible to vaporize the solid source in a neighboring furnace and use a carrier gas to transport the vapors to the wafer. The use of solid sources in thermal diffusion is shown in figure 6.

2.2 Drive-in

Once the dopant atoms have arrived on the wafer surface, they need to be redistributed into the bulk. This process is called *drive-in*. At the same time, the carrier gas could also react with the wafer surface, especially if there is some reactive gas like oxygen (dry ox) or water vapor (wet ox). These could cause oxidation of the Si along with dopant diffusion. While this might be desirable under some circumstances, it would also affect dopant distribution, since the presence of an oxide layer can lead to an increase of *n*-type dopants and decrease of *p*-type dopants, just below the interface. This is shown in figure 7. The drive-in process requires diffusion of the impurities into Si. Depending on the relative size of the impurity atom, this can be either through *vacancy diffusion* or *interstitial diffusion*. The interstitial

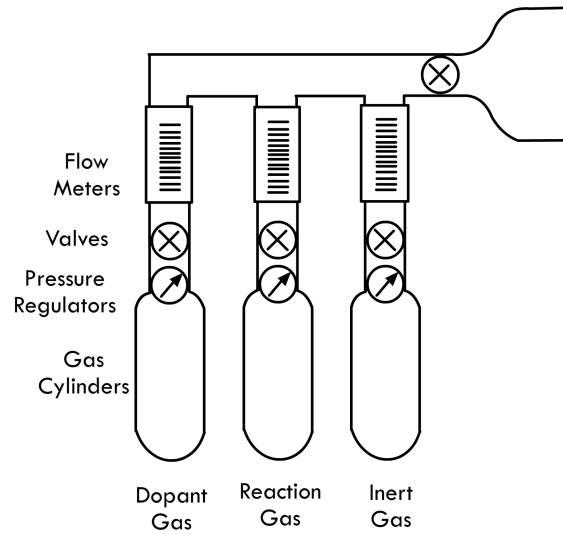


Figure 5: The gas manifold system for a thermal diffusion system. The dopant gas and inert gas are mixed to get the right dopant concentration. There is also a reaction gas, like oxygen, if an oxide layer also needs to be formed. Adapted from *Microchip fabrication - Peter van Zant*.

mechanism is shown in figure 8 while the substitutional mechanism is shown in figure 9. Boron and Phosphorus are small and diffuse by a dual (vacancy and interstitial mechanism) while As and Sb predominantly diffuse by the vacancy mechanism.

3 Diffusion concentration gradients

The amount of impurities that can be incorporated in Si depends on the solid solubility. This depends on the impurity atom and temperature, given by figure 10. To calculate the concentration of the impurities as a function of depth from the wafer surface, Fick's laws of diffusion can be used. *Fick's first law* is written as

$$J = -D \frac{\partial c(x, t)}{\partial x} \quad (1)$$

where J is the flux of impurity atoms, which is a constant with respect to time (steady state diffusion) and $c(x, t)$ is the concentration at depth x and time t and D is the diffusion coefficient. D is temperature dependent and is

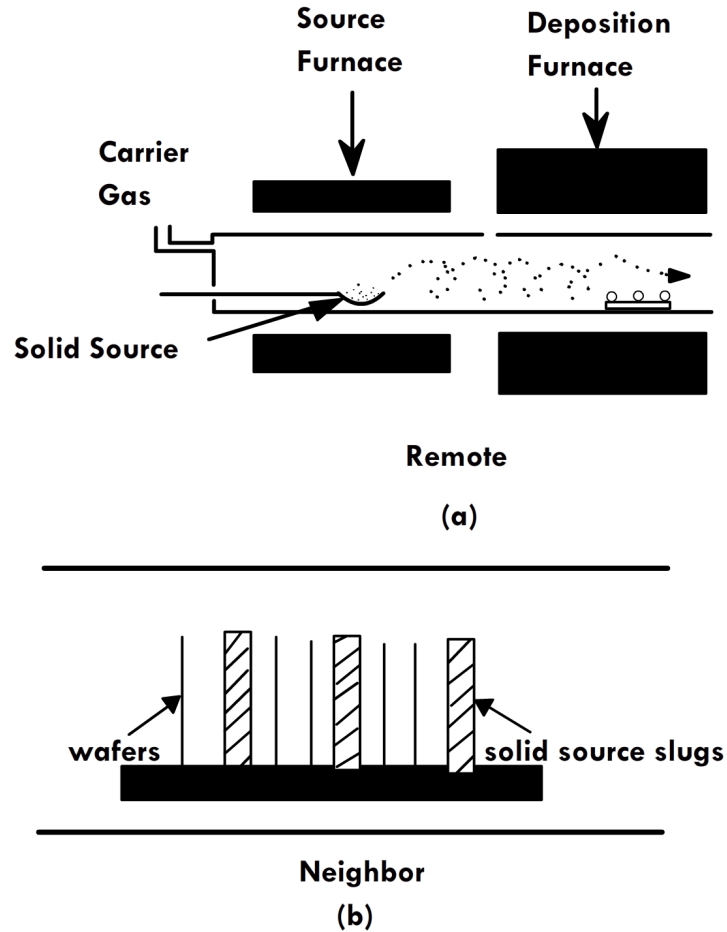


Figure 6: Solid sources for thermal diffusion can be either (a) remote or (b) neighbor sources. In a remote source the solid is vaporized and the vapors are passed into the furnace. In neighbor sources, the solid is loaded in the furnace along with the wafers. Both sources produce different concentration profiles in the wafer. Adapted from *Microchip fabrication* - Peter van Zant.

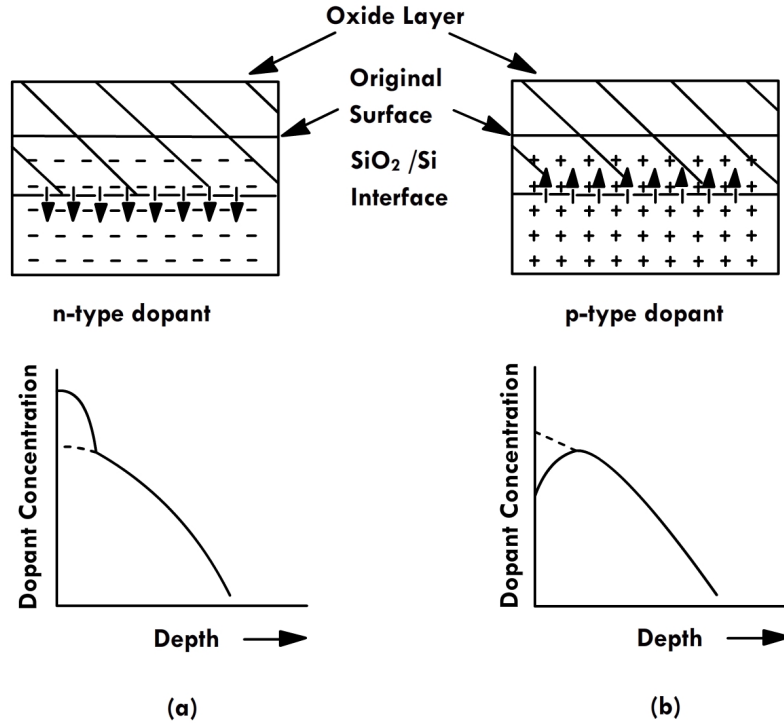


Figure 7: Growth of oxide layer on Si can cause (a) pile-up of n type impurities and (b) depletion of p type impurities. The oxide layer can be grown along with dopant diffusion or once diffusion is complete. Adapted from *Microchip fabrication* - Peter van Zant.

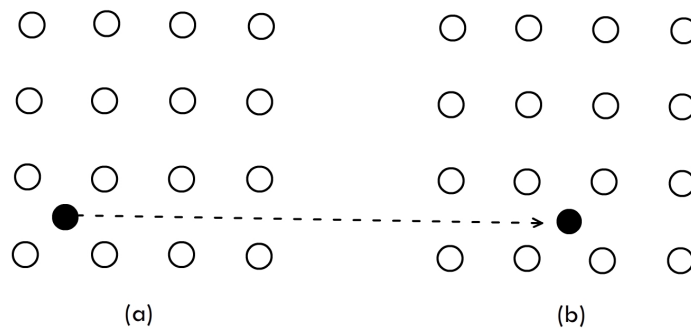


Figure 8: Interstitial diffusion mechanism showing motion of dopant atom from position in (a) to (b). Here, the dopant atom is smaller than the silicon atom, so that interstitial doping is possible. Adapted from *VLSI fabrication principles* - S.K. Ghandhi

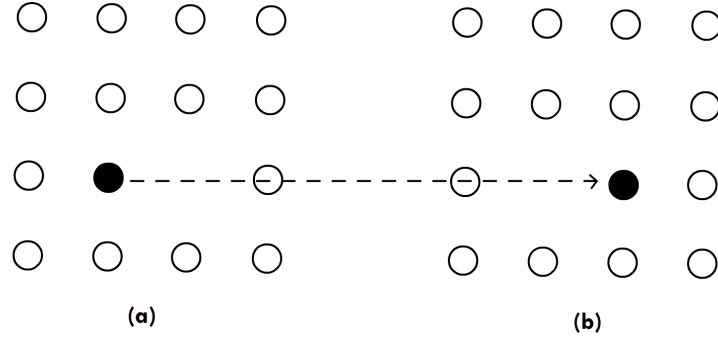


Figure 9: Substitutional diffusion mechanism showing motion of dopant atom from position in (a) to (b). Substitutional diffusion happens when the dopant size is comparable to the Si atom. Adapted from *VLSI fabrication principles* - S.K. Ghandhi

given by

$$D = D_0 \exp\left(-\frac{E_a}{k_B T}\right) \quad (2)$$

where D_0 is the pre-exponent factor and E_a is the activation energy for diffusion. For unsteady state, with the flux varying with time and position, a more general equation is *Fick's second law*, given by

$$\frac{\partial c(x, t)}{\partial t} = -\frac{\partial J(x, t)}{\partial x} = D \frac{\partial^2 c(x, t)}{\partial x^2} \quad (3)$$

The assumption is that D is not a function of concentration. By applying equations 1 and 3, it is possible to calculate the concentration gradient under various conditions. The maximum amount of dopants that can be incorporated is given by the impurity solubility shown in 10. This represents the thermodynamic limit. There are two common doping conditions that exist in thermal diffusion.

3.1 Constant surface concentration

For gaseous and liquid sources, there is a constant concentration of impurities at the surface. There is a vapor of impurity atoms (it is also true for a solid source with remote evaporation) that maintains the constant concentration on the surface. Also, the diffusion length is much smaller than the wafer thickness so that the wafer can be approximated as a semi-infinite solid. In

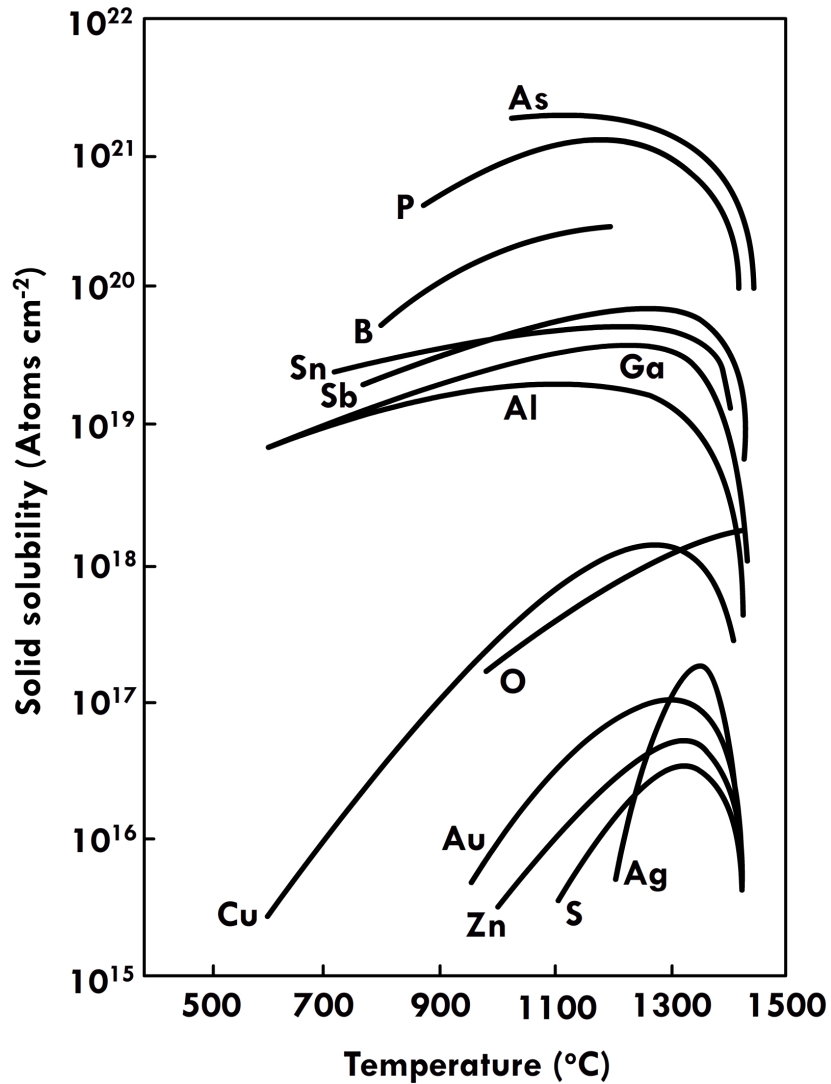


Figure 10: Solid solubility of various impurities in Si as a function of T. Commonly used *p* and *n* dopants have high solubility, approaching the level of degenerate semiconductors. Other impurities like metals and oxygen have solubility levels of a few *ppm*. Adapted from *Microchip fabrication* - Peter van Zant.

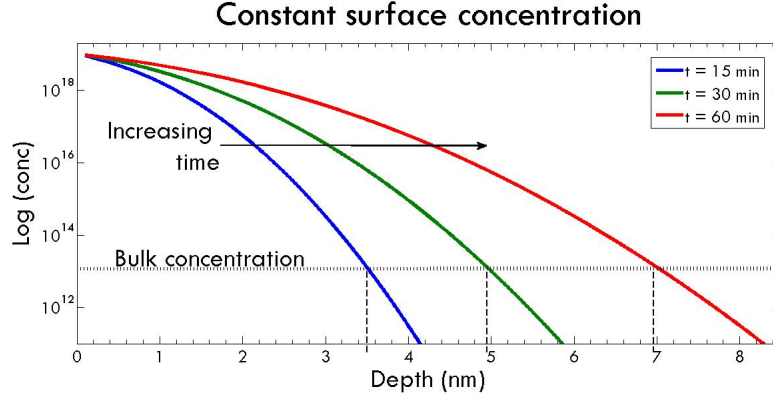


Figure 11: Concentration profiles for constant surface concentration with increasing time. The maximum concentration is at the surface. With increasing time, the junction depth goes deeper within the wafer. The junction depth is defined as when the dopant concentration becomes equal to the wafer bulk doping level. In this case, the bulk dopant concentration is 10^{13} cm^{-3} . The plot was generated in MATLAB.

this case, the impurity concentration, $C(x, t)$, is given by

$$C(x, t) = C_s \operatorname{erfc}\left(\frac{x}{2\sqrt{Dt}}\right) \quad (4)$$

where C_s is the surface concentration and erfc is the complementary error function.

Consider an example of diffusion under a constant surface concentration of 10^{19} cm^{-3} and a bulk dopant concentration of 10^{13} cm^{-3} . The wafer temperature is 1000 K , and the diffusion coefficient at this temperature is $2.9 \times 10^{-22} \text{ m}^2 \text{ s}^{-1}$. The error function solution, for this system, is plotted in figure 11, for three different times, 15, 30, and 60 minutes. Equation 4 can also be used to calculate the junction depth for a pn diode. If the base wafer is p type with concentration C_p , the junction depth (where electron and hole concentration is the same) by n type impurity diffusion is given by

$$\frac{C_p}{C_s} = \operatorname{erfc}\left(\frac{x_{pn}}{2\sqrt{Dt}}\right) \quad (5)$$

where x_{pn} is the junction location. With increasing time, the junction depth increases.

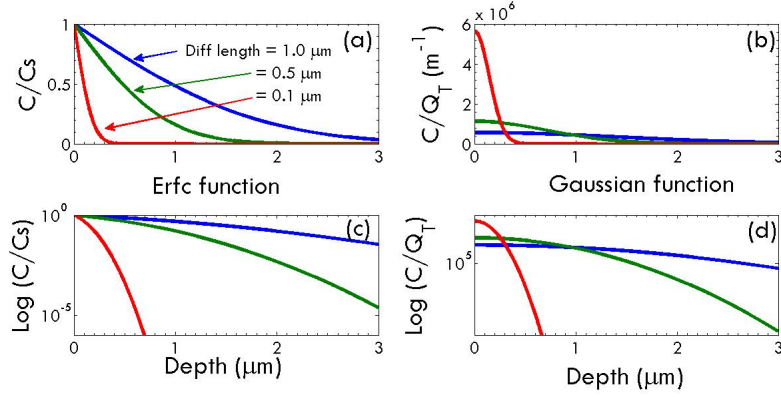


Figure 12: Comparison of (a) and (c) error function solution and (b) and (d) Gaussian solution to the diffusion equations, for different diffusion lengths. Compared to the error function solution, the slope of the Gaussian solution is less steep. Thus, there is greater penetration of the dopants in the bulk, with respect to the surface concentration, for the same diffusion length. Plot was generated using MATLAB.

3.2 Constant total dopant

In this scenario, the total amount of dopant atoms at the start of the diffusion process is a constant and the concentration of the atoms at the surface gradually decreases with time. This happens in the case of doping from a solid source that is placed close to the wafer (either solid slugs close to the wafer or dopants spun on the wafer surface). Let Q_T be the total amount of dopants on the surface (at start of diffusion) per unit area. Then, the concentration, $C(x, t)$, is given by

$$C(x, t) = \frac{Q_T}{\sqrt{\pi D t}} \exp\left(-\frac{x^2}{4 D t}\right) \quad (6)$$

This is a Gaussian function, in contrast to the error function solution for a constant surface concentration. The change in surface concentration as a function of time can be obtained from equation 6 with $x = 0$. This gives

$$C(0, t) = \frac{Q_T}{\sqrt{\pi D t}} \quad (7)$$

The error function and Gaussian solutions are compared in figure 12.

Table 2: Diffusion parameters for different impurities in Si

Parameter	B	P	As	Sb
D_0 ($cm^2 s^{-1}$)	0.76	3.85	24	0.214
E_a (eV)	3.46	3.66	4.1	3.65

3.3 Diffusion example

Consider a n -type Si wafer with As concentration of $10^{16} cm^{-3}$. A pn junction is to be formed by diffusing p -type impurity, using a solid boron source. The diffusion is to be carried out at 1000 K . The diffusion coefficients for various dopants in Si, as a function of temperature, are shown in figure 13. Extrapolating from figure 13, the diffusion coefficient of B in Si at 1000 K is $2.9 \times 10^{-18} cm^2 s^{-1}$. The diffusion coefficient can also be calculated from known values of pre-exponent factor (D_0), which is $0.76 cm^2 s^{-1}$, and activation energy, which is (E_a) is 3.46 eV for B in Si, using equation 2.

Since B is a solid source, this is a case of constant total dopant concentration, discussed in section 3.2. Hence, the concentration as a function of depth and time, $C(x, t)$, is given by equation 6. The term $\sqrt{4Dt}$ represents the diffusion length for two-dimensional diffusion. Let Q_T be $10^{13} atoms cm^{-2}$, is the total surface concentration per unit area at the start of the process. Using equation 6, it is possible to calculate the junction depth after a certain time, say $t = 2 hrs$. Junction depth, x_{pn} , is defined at the depth when the n and p concentrations are equal. At 2 hrs, and at 1100 $^{\circ}C$, the depth, x_{pn} is calculated to be 8.3 nm, represents a *shallow junction*.

To increase the junction depth (for the same time), the diffusion coefficient has to be increased. One way to achieve this, is by increasing temperature. For a temperature of 1200 K , the value of D is $2.3 \times 10^{-15} cm^2 s^{-1}$. This translates, using equation 6, into a junction depth of 180 nm.

A combination of error function and Gaussian diffusion profiles is used for forming transistor junctions, e.g. BJT junction shown in figure 1. In this case, the first layer is formed by diffusion from a solid source (Gaussian profile) followed by diffusion from liquid or gas source (erf profile), as shown in figure 14. The intersection points for the two concentration profiles and the point where the Gaussian profile intersects the base dopant concentration, represent the junction locations. This is shown in figure 15. The dopant profiles will get affected with any subsequent annealing steps. Pre-exponent factors and activation energies for different dopants in Si are listed in table 2.

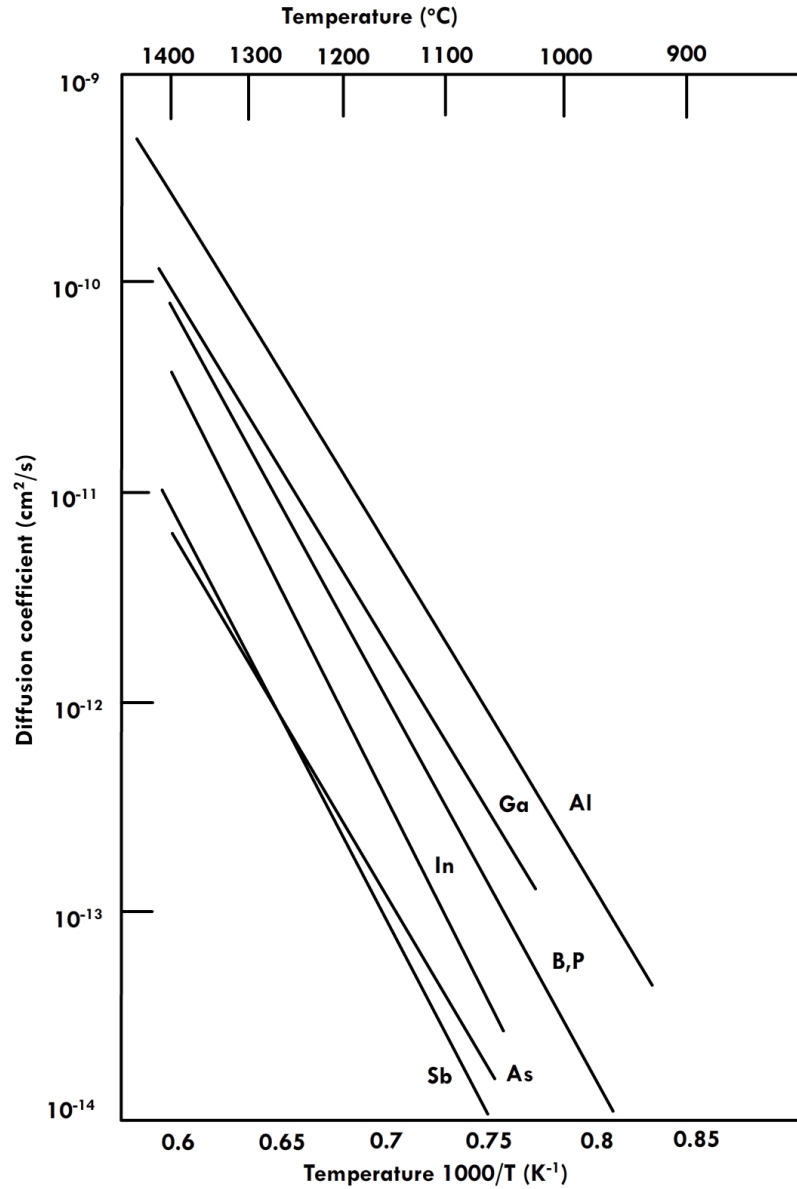


Figure 13: Diffusion coefficients as a function of temperature for different impurities in Si. This is a semi log plot of concentration vs. inverse temperature and the slope gives the activation energy. Adapted from *VLSI fabrication principles* - S.K. Ghandhi

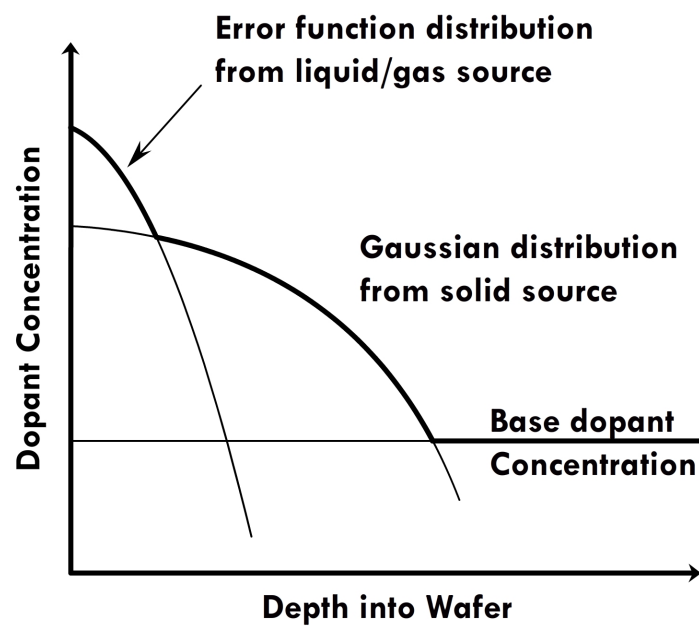


Figure 14: Concentration profiles for multiple diffusion sources. The Gaussian profile is less steep than the error function solution. The intersection of the two profiles and the intersection of the Gaussian profile with the bulk doping level represent the positions of the junctions. Adapted from *Microchip fabrication* - Peter van Zant.

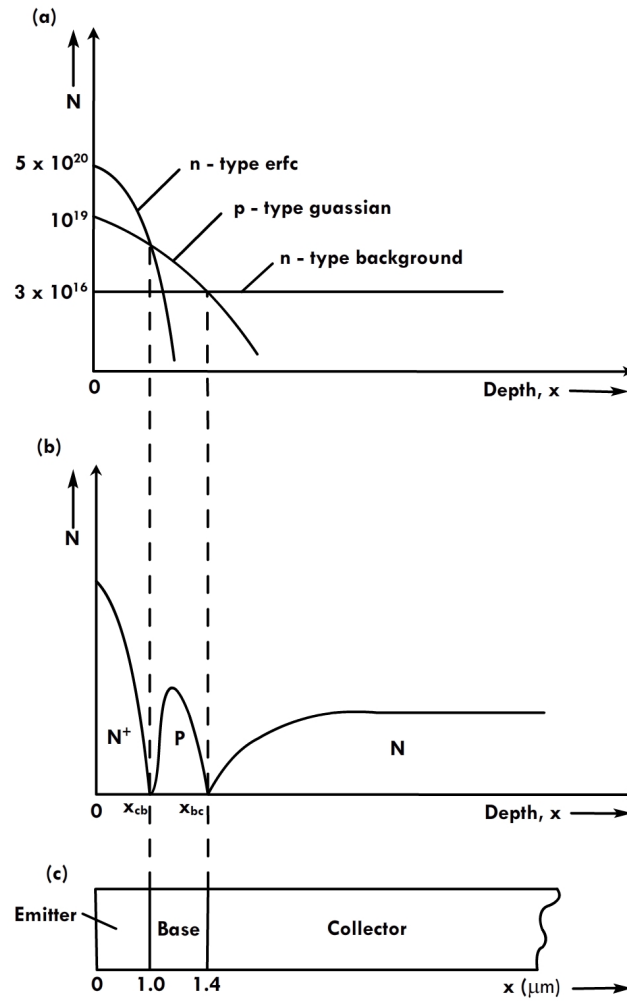


Figure 15: (a) Concentration profiles from a solid and liquid/gas source (b) Net dopant concentration as a function of depth (c) Formation of a BJT resulting from the dopant concentration in (b). Adapted from *VLSI fabrication principles* - S.K. Ghandhi

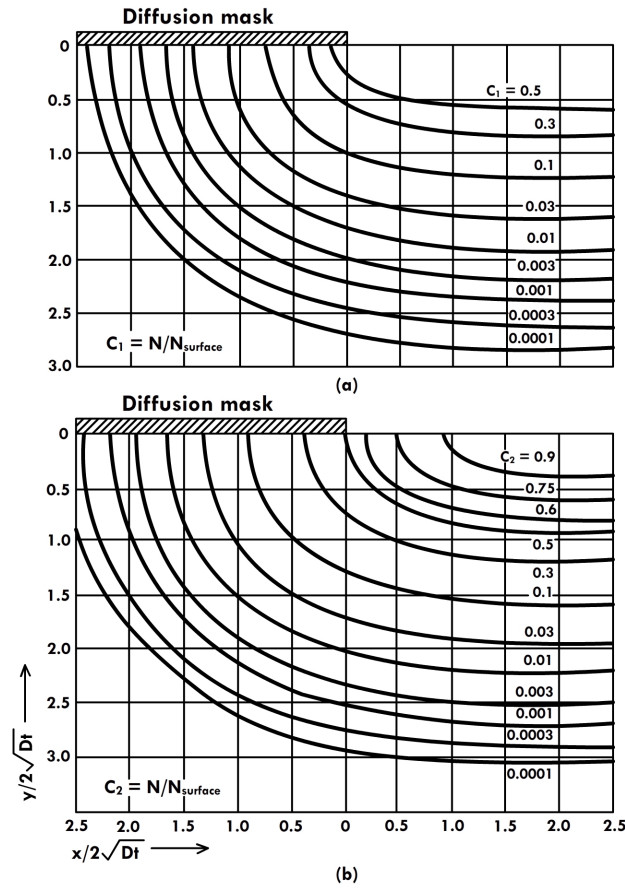


Figure 16: Lateral diffusion below an oxide window for (a) constant source diffusion (b) depleting source diffusion. Lateral diffusion increases with diffusion length, either increase in D or t . Adapted from *VLSI fabrication principles* - S.K. Ghandhi

4 Ion implantation

The biggest limitation of thermal diffusion is that the process is isotropic i.e. lateral diffusion cannot be avoided, though diffusion coefficients in different crystallographic directions might be different. Thus, an oxide window that serves as a mask to protect certain regions of the wafers, can be ineffective due to lateral diffusion. This is shown in figure 16. This is especially important for doping small regions (due to device miniaturization). Doping control is also difficult to achieve due to presence of concentration gradients. These gradients will change in subsequent annealing steps. Thus, there is a thermal budget associated with doping.

Ion implantation is a relatively newer doping technique that operates close to room temperature. It is a *physical process* of doping, not based on a chemical reaction. Since ion implantation takes place close to room temperature, it is compatible with conventional lithographic processes, so small regions can be doped. Also, since temperature is low, lateral diffusion is negligible.

In ion implantation, dopant atoms are ionized, isolated, accelerated and made to impinge on the wafer surface. These atoms penetrate some depth into the material and get embedded into the wafer. The schematic of the process is shown in figure 17. The source material is usually in the form of a gas e.g. AsH_3 , PH_3 , and BF_3 are some common sources. Similarly, elemental sources like As and P are also used as solid sources. Electron bombardment is used to create ions. The ions are then separated using a *mass analyzer*, which is a 90° magnet, which bends the ions depending on the mass. After selection, the desired ions are then accelerated and made to impinge on the wafer surface. Beam scanning or *rastering* is also possible using electric field coils to deflect the ion beams.

The penetration depth of the ions depend on their energy (changed by the accelerating field). The concentration profile for ion implantation is shown in figure 18. The maximum concentration is at a certain depth below the surface, called **range**. In thermal diffusion, the maximum concentration is at the surface and the concentration decreases with depth. The range depends on the ion type and the energy, as shown in figure 19. There are two stopping mechanisms - the nucleus of the wafer atoms and the interaction of the positive ions with the electrons. In ion implantation, the *beam density* ($\# \text{ ions/cm}^2$), *ion energy*, and *orientation of the wafer matter*.

In ion implantation, since the wafer surface is impacted by high energy ions, it can cause damage by knocking Si atoms from their position, causing *local structural damage*. This needs a post thermal annealing treatment to repair the damage. There are two ways of doing this.

1. Tube furnace - low temperature annealing (600-1000 $^\circ\text{C}$). To minimize lateral diffusion.
2. Rapid thermal annealing - higher temperatures are possible but for shorter times.

Ion implantation is especially useful with device scaling. It can be used to create *shallow junctions*, by having a small range. It can also be used to dope small regions. It is usually used later in the process flow when thermal budgets are tight and the high temperature of thermal diffusion is not allowed.

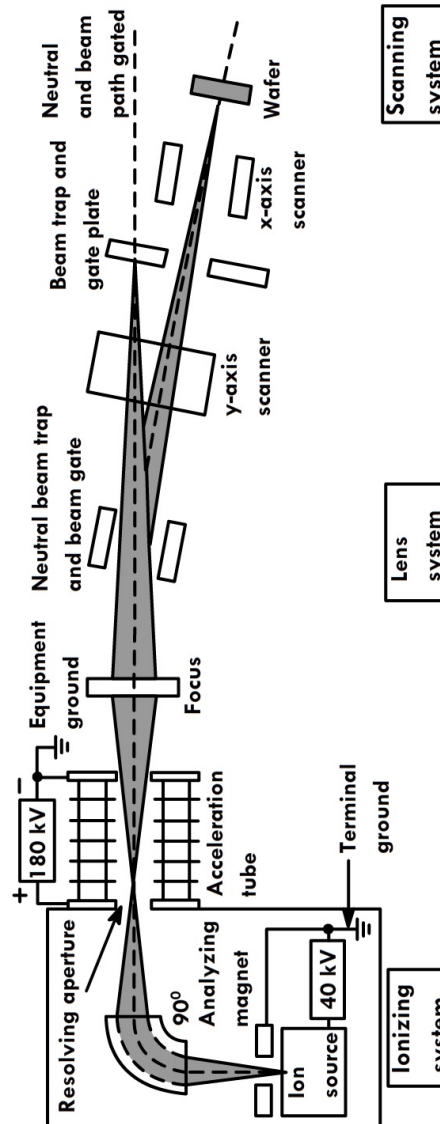


Figure 17: Schematic of the ion implantation process. Dopant atoms are ionized by bombarding with electrons. These are then isolated, accelerated, and then impinged on the wafer. There is also a scanning system that allows the ion beam to scan over the wafer surface. Adapted from *Fundamentals of semiconductor manufacturing and process control* - May and Spanos.

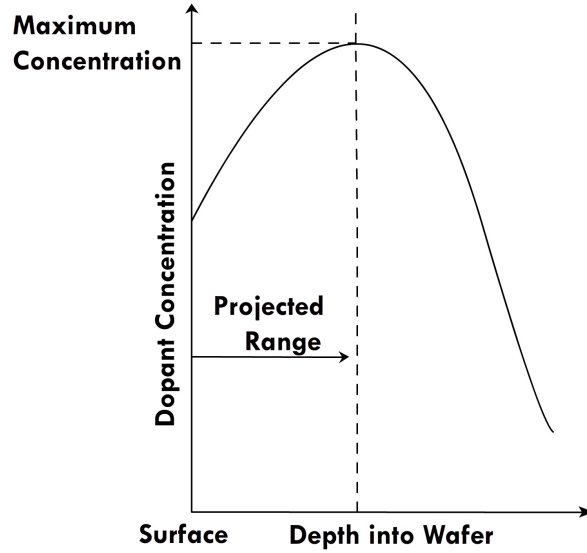


Figure 18: Concentration profile for ion implantation. The maximum concentration is below the wafer surface, unlike thermal diffusion, and the depth increases with the ion energy. Adapted from *Microchip fabrication* - Peter van Zant.

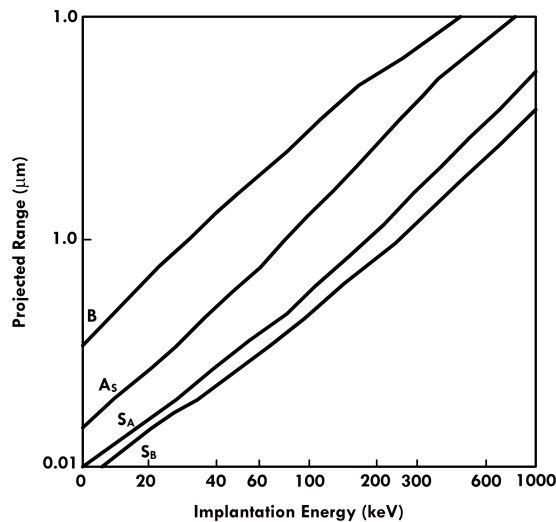
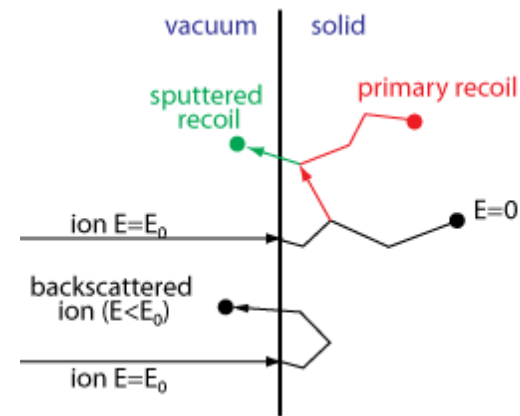
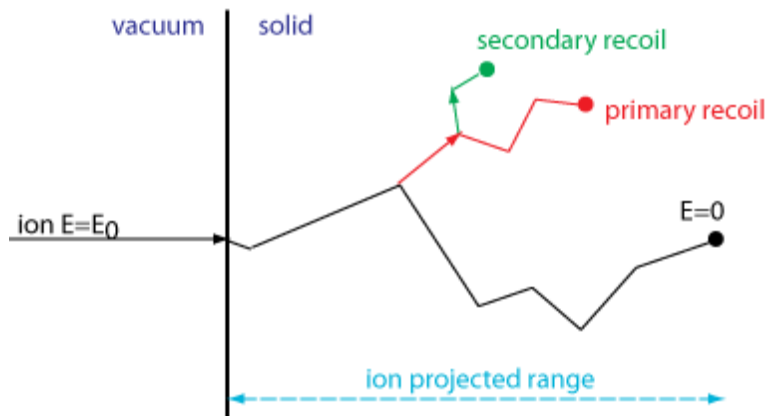


Figure 19: Plot of the range (penetration depth) vs. ion energy for three different dopants, i.e P, As, and S. Channeling effects are shown for S, marked S_A and S_B . Adapted from *Microchip fabrication* - Peter van Zant.

What is ion implantation, in short ?

- bombardment of a solid target with an energetic **ion beam** (typically **1keV - 1MeV**) $\Rightarrow$ ions penetrate through the material, slow down by transferring their energy to the target atoms/electrons and eventually stop at a certain depth (**$\sim 0-1\mu\text{m}$**) below the surface



- It is basically a **surface** or **near-surface** processing technique. Due to their stronger interactions with target atoms, ions penetrate much less in matter than electrons, for the same energy.

Projected ranges of charged particles - electrons, protons, alpha particles, heavy ions - in **Silicon**

<i>particles</i>		<i>energy (MeV)</i>	<i>range (μm)</i>
electrons (e^-)	(1)	2	~ 0.5 cm
protons (H^+)	(1)	2	~ 50 μm
alpha (He^{++})	(1)	2	~ 7 μm
B ions	(2)	2	~ 3 μm
P ions	(2)	2	~ 2 μm
As ions	(2)	2	~ 1.3 μm

EM radiation (**X-** or **γ -Rays**) and neutral particles (**neutrons**) interact much less with target atoms and have much longer (> cm) penetration ranges → *bulk* process.

Sources:

(1) <http://www.nist.gov/pml/data/star/> (NIST Stopping Power and Range Tables)

(2) <http://www.cleanroom.byu.edu/rangestruggle.phtml> (BYU - Brigham University)

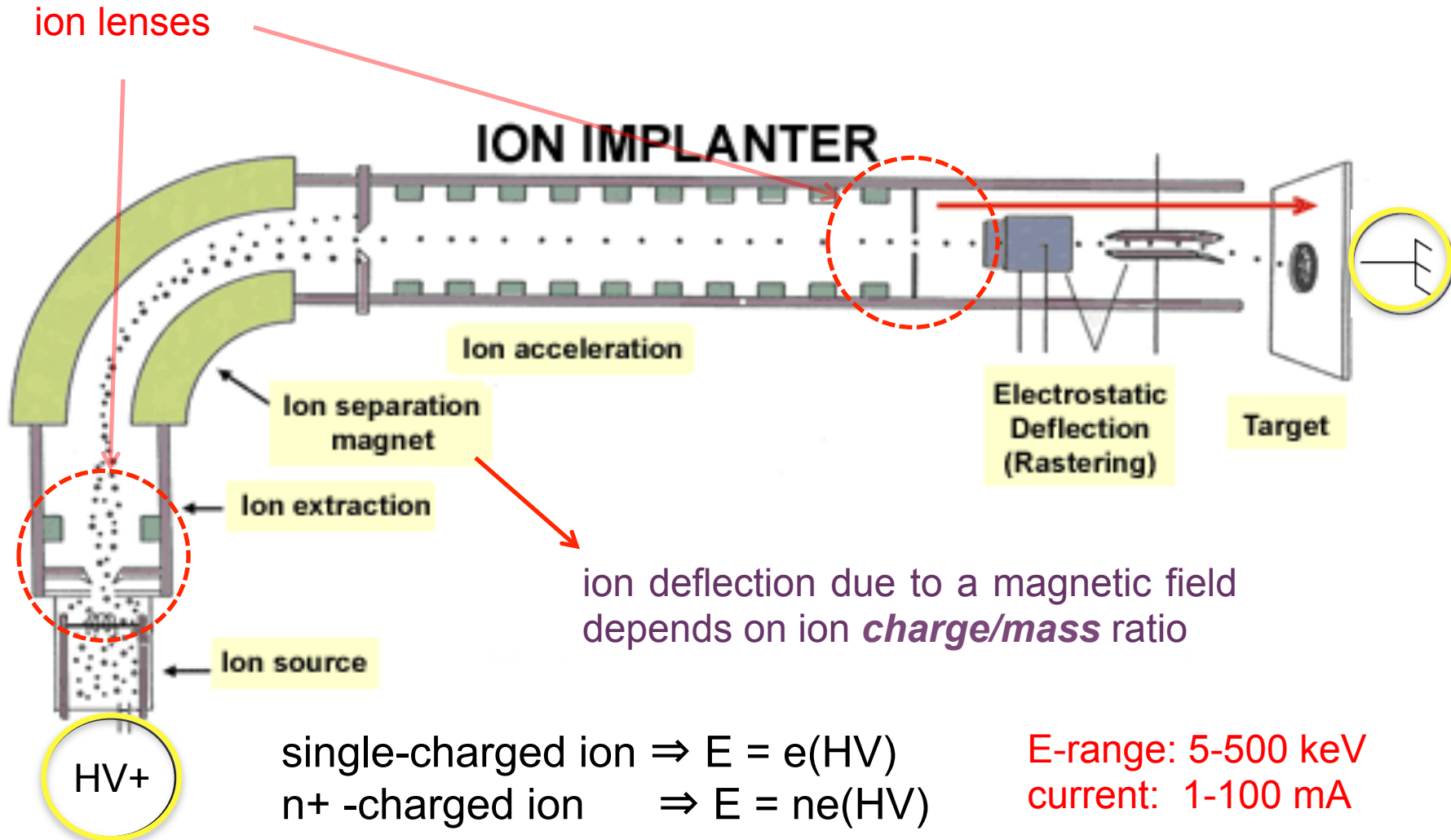
Why ion implantation ?

- To modify/improve properties (physical, chemical) of materials in a thin (**50nm-1 μ m**) surface layer
- Changes occur because (main reasons):
 - 1) we implant chemical species different from the target, which, although in a small total amount, may still reach high relative concentrations on a microscopic scale (***chemical modification***)
 - 2) ion irradiation damages the structure of the target, generating radiation defects, changes of phase, atom removal, etc. (***structural modification***)

What do we need to perform ion implantation ?

- Produce a beam of ions (***ion source***)
- Select among beam atoms just the chemical species we want to implant (***magnetic mass filter***)
- Accelerating ions at the desired energy (***acceleration column***)
- Bring the ion beam to hit the surface of the target material (***beamline: ion lenses, beam scanner, neutral trap,...***)

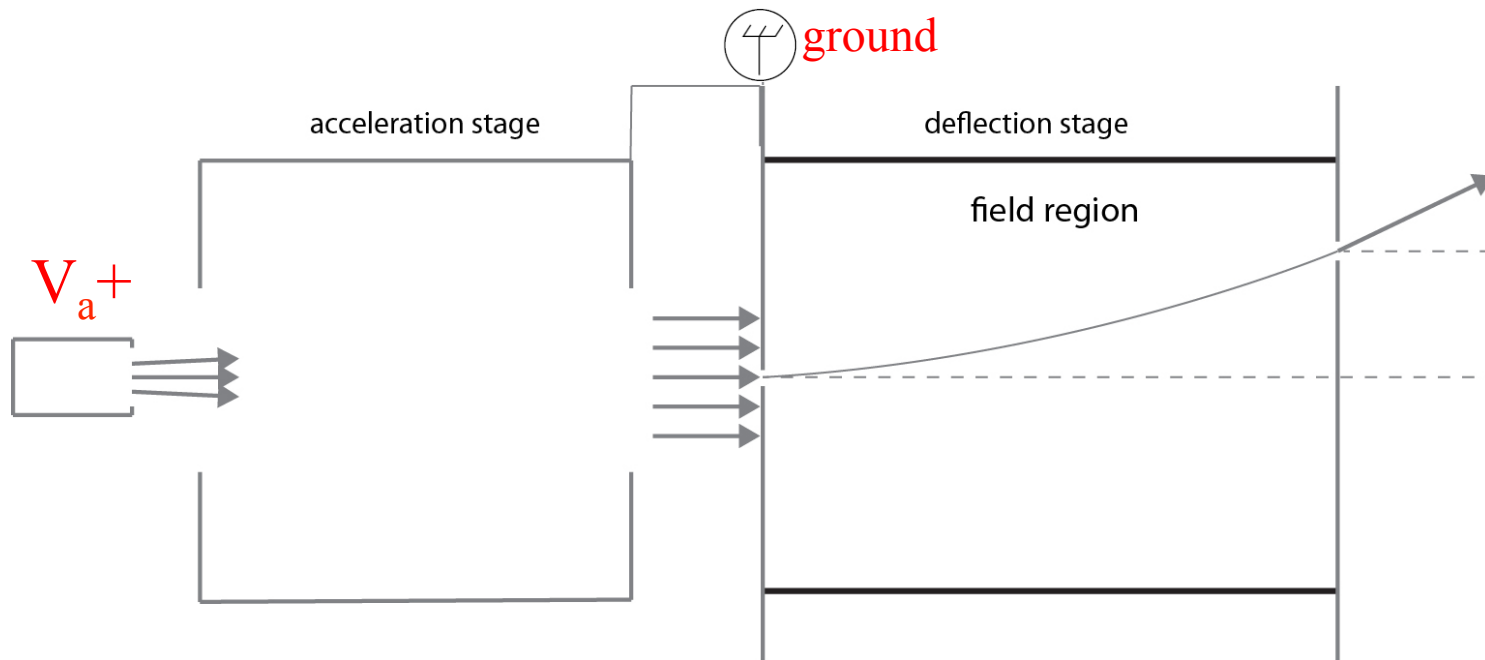
Schematics of a *single-ended* ion implanter



Ion beam manipulation by electromagnetic fields

- force acting on an ion of positive charge q and velocity v in an electromagnetic field:

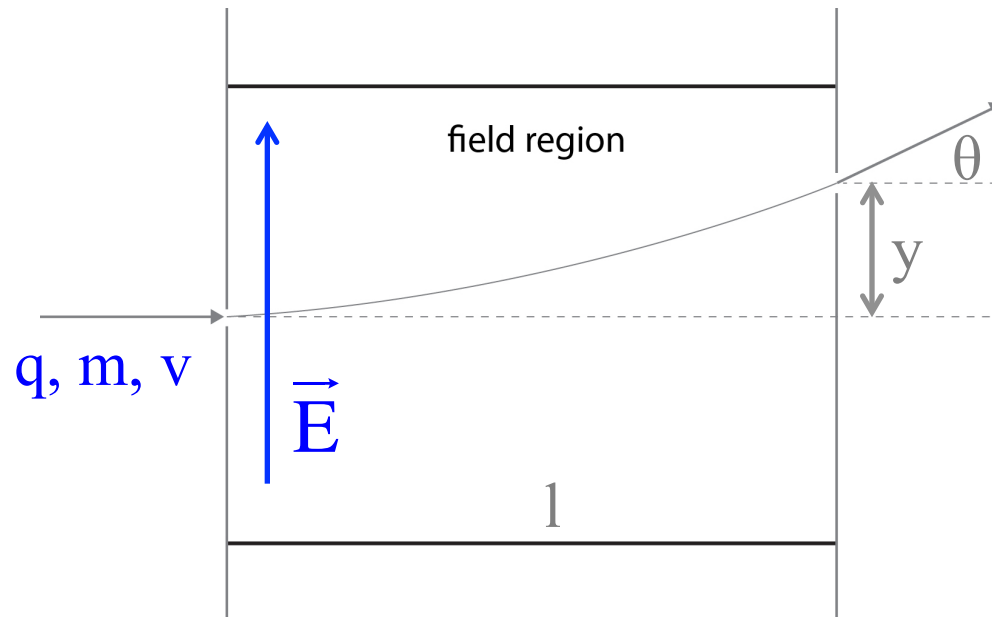
$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B} \quad (\text{electrostatic} + \text{Lorentz force})$$



Ion beam manipulation by electromagnetic fields

■ Pure electrostatic

$$\vec{F} = q\vec{E}$$



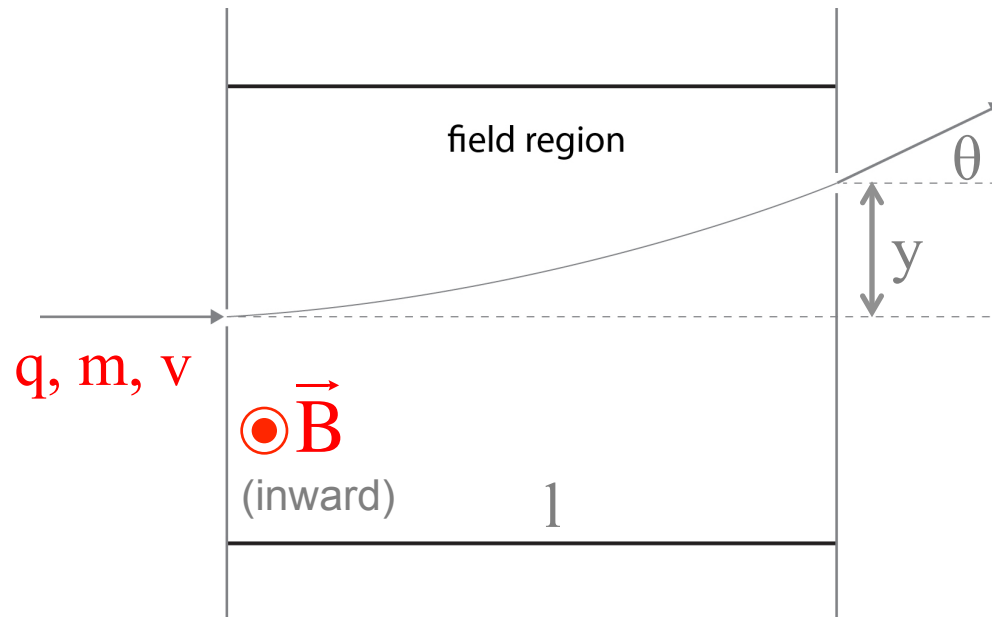
$$\vartheta \approx \frac{E}{2V_a} l$$

ion deflection independent of ion mass or ion charge

Ion beam manipulation by electromagnetic fields

■ Pure magnetic

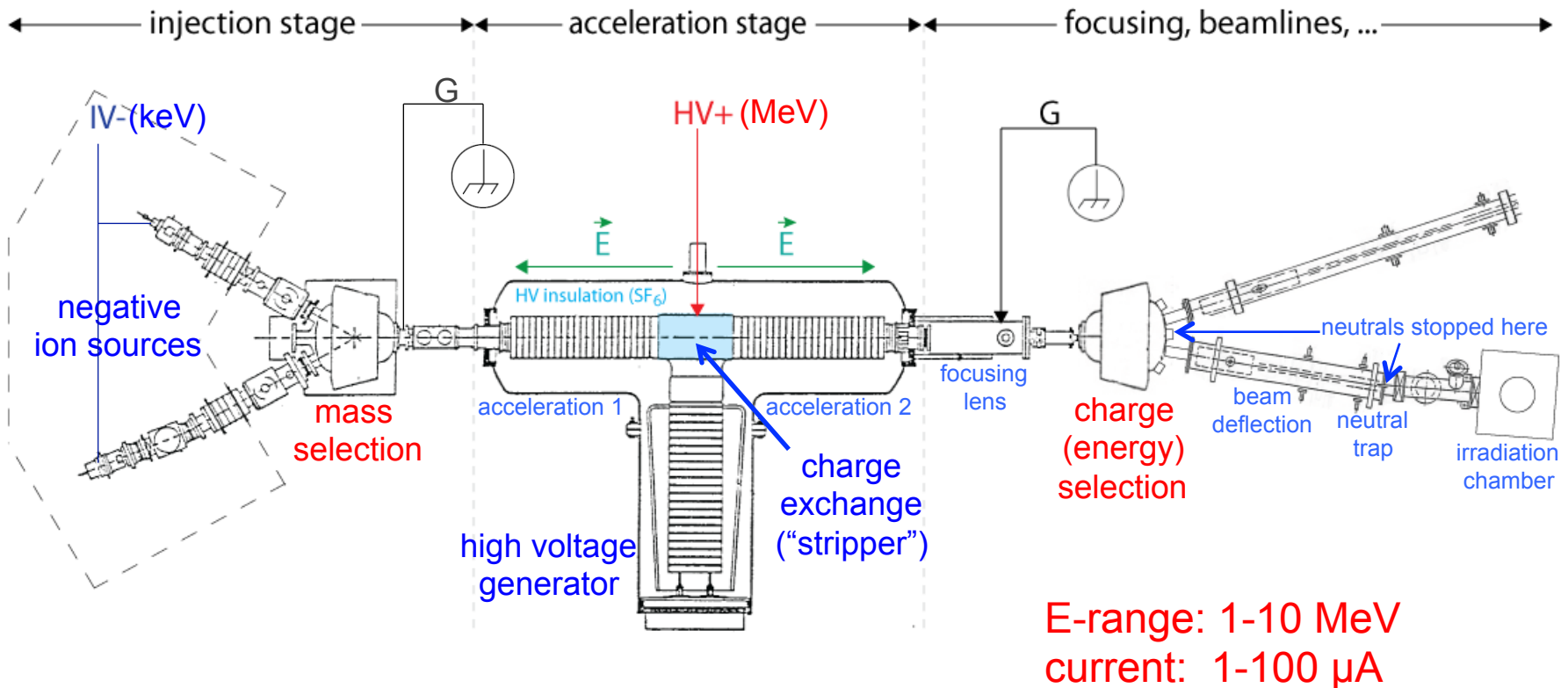
$$\vec{F} = q\vec{v} \times \vec{B}$$



$$\vartheta \simeq \left(\frac{q}{m} \right) \frac{B}{v} l$$

for the same velocity/energy (not changed by the magnetic field) deflection depends on **charge to mass ratio**

Schematics of a tandem (double-ended) ion implanter



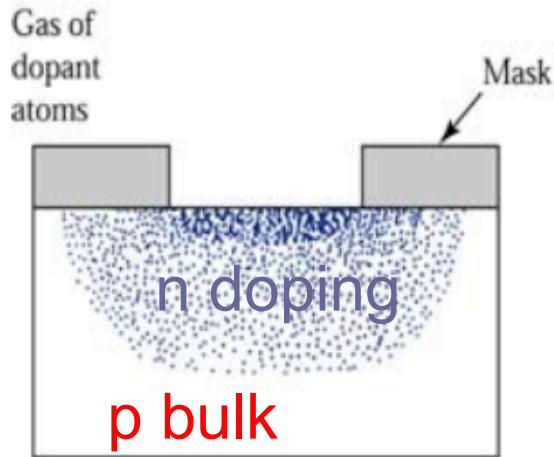
$$\text{single-charged ion} \Rightarrow E = (IV) + e(HV) + e(HV) = IV + 2e(HV)$$

$$n - \text{stripped ion} \Rightarrow E = (IV) + e(HV) + ne(HV) = IV + (n+1)e(HV)$$

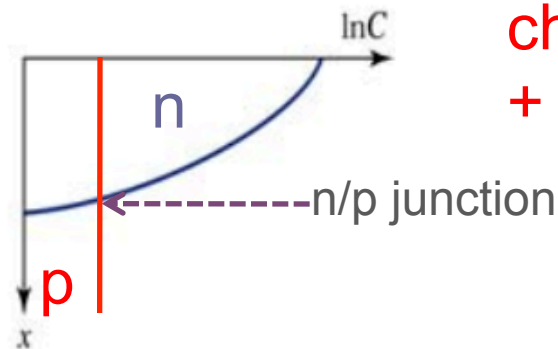
Summary of electric and magnetic fields for ion beam manipulation

electric field (deflection independent on mass and charge)	magnetic field (deflection dependent on the charge/mass ratio)
accelerate ions to reach desired beam energy	mass filter : select ion mass (for ions of the same charge)
optics: focus ion beams across the implanter column to optimize beam transmission	energy filter : select ion charge (for ions of the same mass)
scan a small beam to cover a large irradiation area	
deflect the beam to separate charged from neutral particles (<i>neutral trap</i>)	

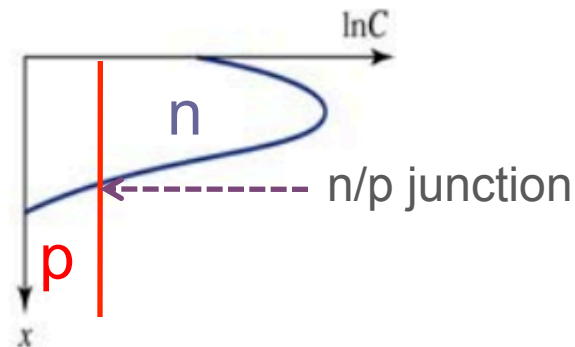
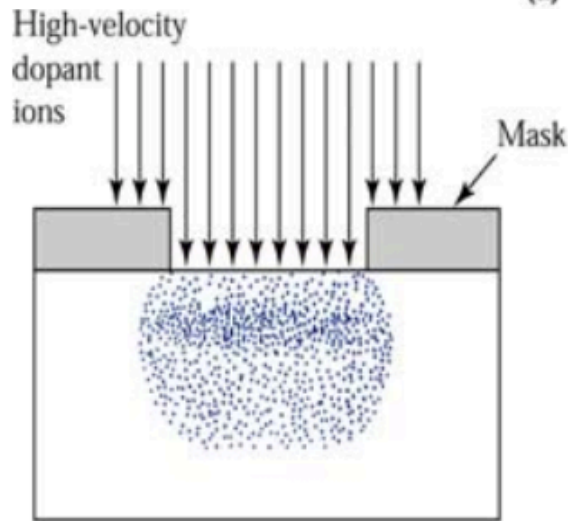
Different techniques for *electrical doping* of semiconductors



(a)



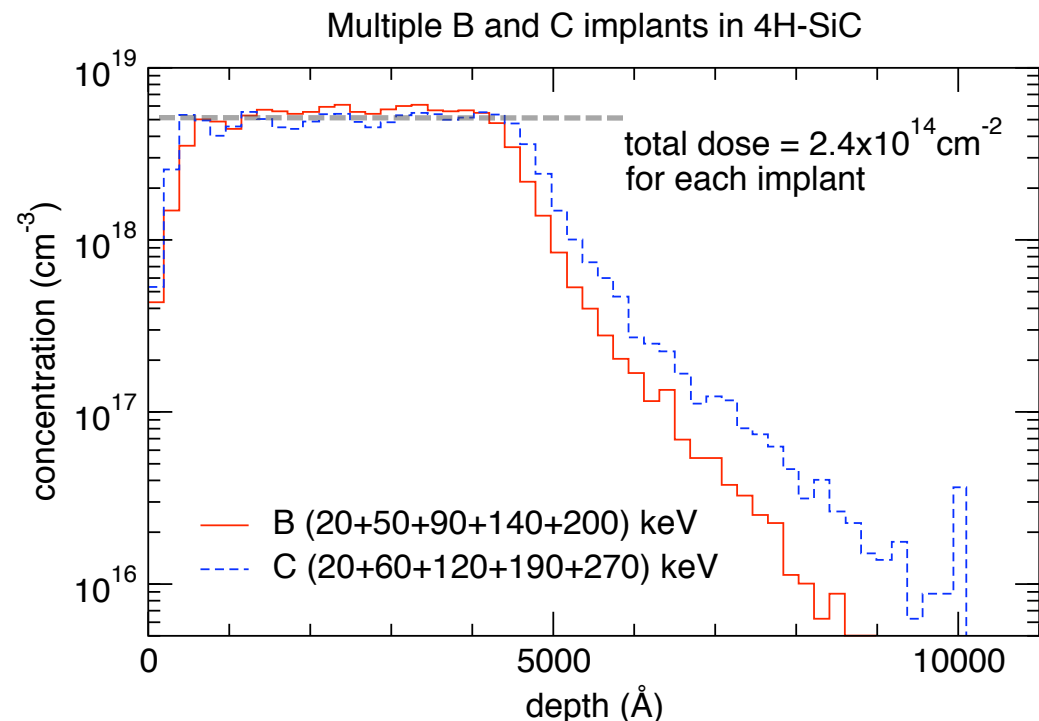
chemical deposition
+ thermal “drive-in”



ion implantation

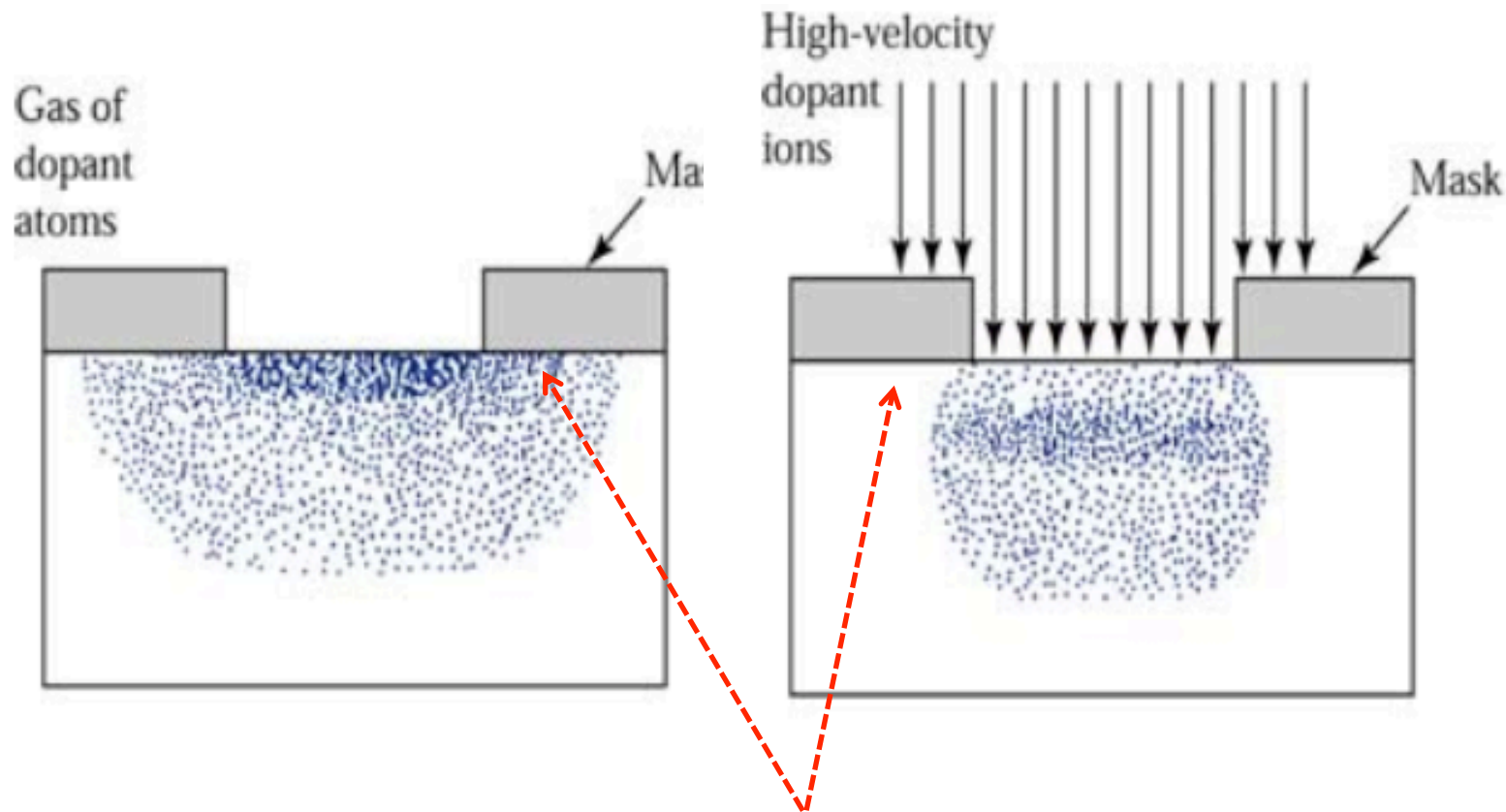
Ion implantation: **PROS**

- Good control ($\leq 10\%$) of the **fluence** (ions cm^{-2}), or total amount of implanted atoms
- It's a *cold* process: very limited amount of thermal **diffusion** during implantation $\Rightarrow$ accurate control of depth distribution of ions.
- using **multiple implants** at different fluences, one has large degrees of freedom in designing ion distribution profiles that would be impossible by chemical deposition + drive in (*example on the left: multiple implantation in 4H-SiC – BCA simulation*)



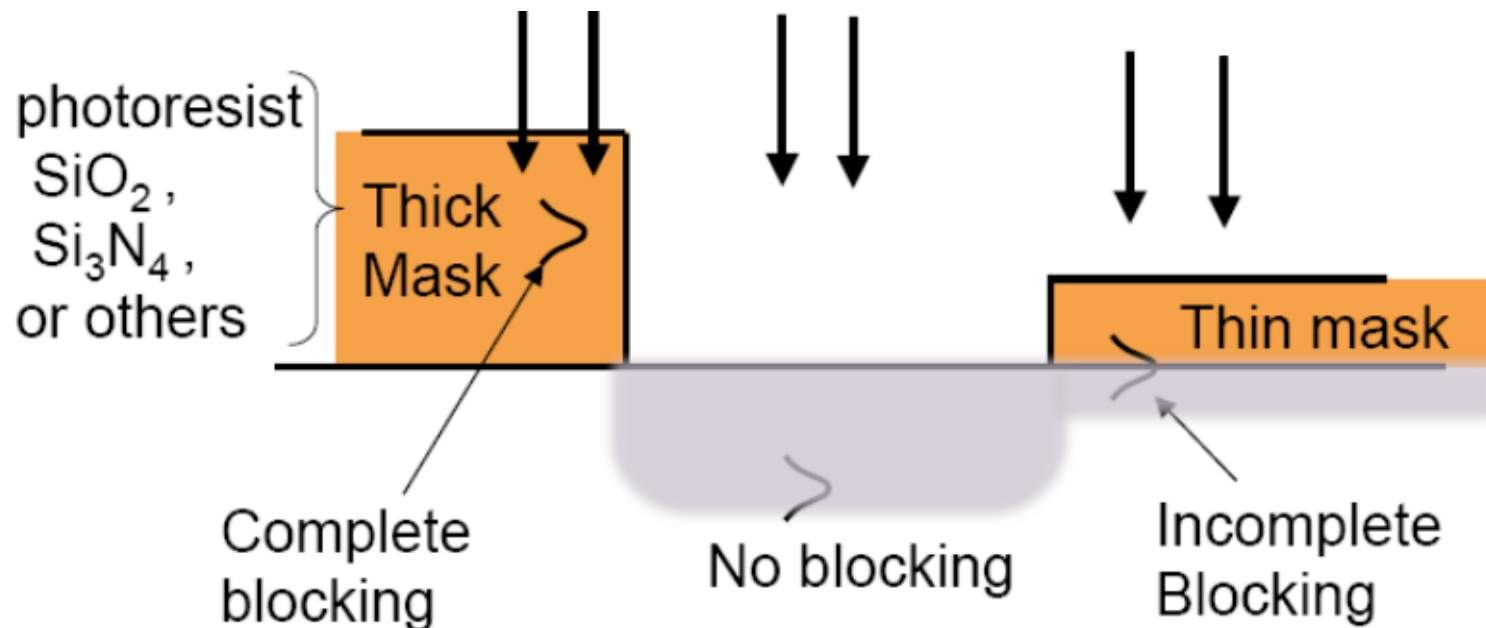
Ion implantation: **PROS**

- Reduced *lateral* (under-mask) penetration of dopant

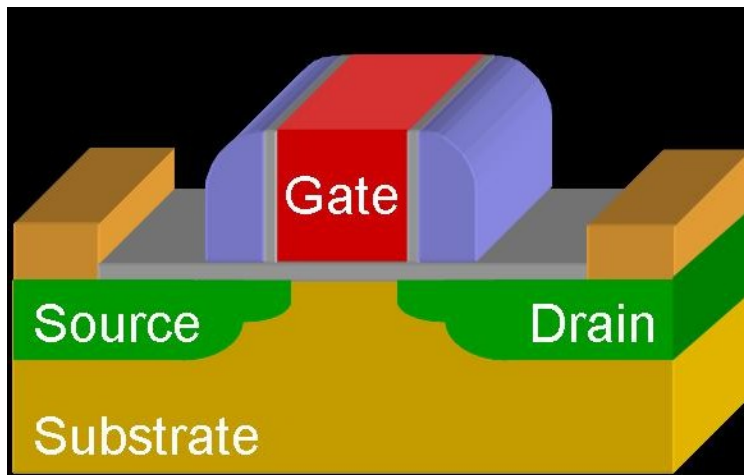
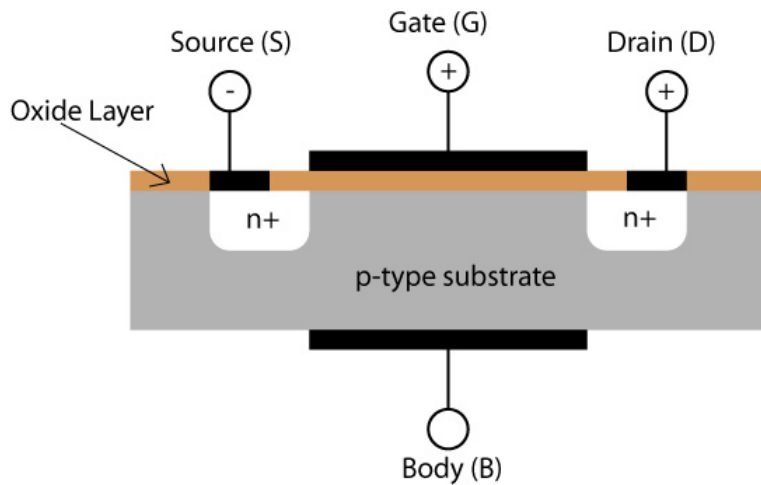


Ion implantation: **PROS**

- **ion masking** allows *lateral selectivity* and *in-depth modulation* of ion distribution in a single implant step



A simple microelectronic device: the **MOSFET**

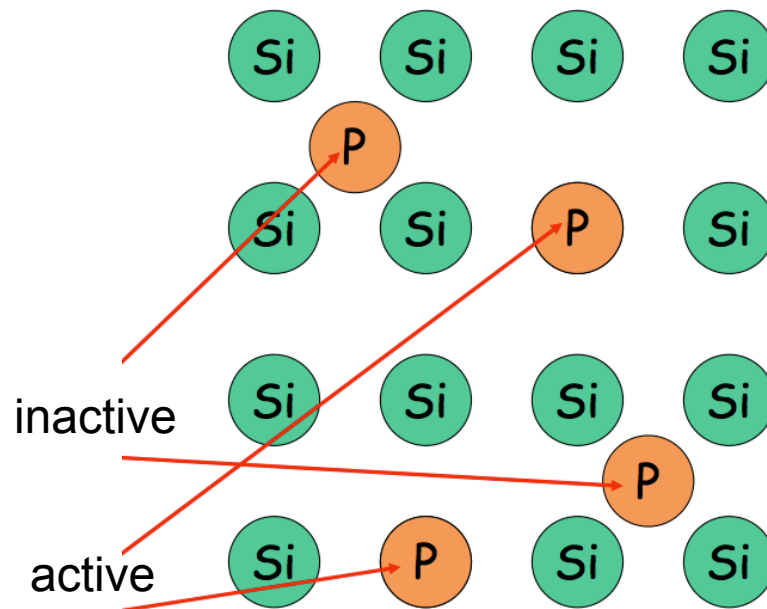


The sequence of many processes like: *epitaxial growth, film deposition, masking, selective etching, ion implantation, oxidation, thermal treatment* ... allows the design and fabrication of complex microelectronic devices in Silicon and other semiconductor materials. For its characteristics, ion implantation is particularly suitable to be integrated in such a complex process flow.

Figures on the left show the schematics of a Metal-Oxide-Semiconductor Field- Effect Transistor (MOSFET), the relatively simple device most widely used in modern digital microcircuits.

Ion implantation: **CONS**

- **Radiation damage:** for many applications such as electrical doping of semiconductors, damage must be recovered by subsequent thermal processing (typically in the range 500-900 C for 5' – 60')

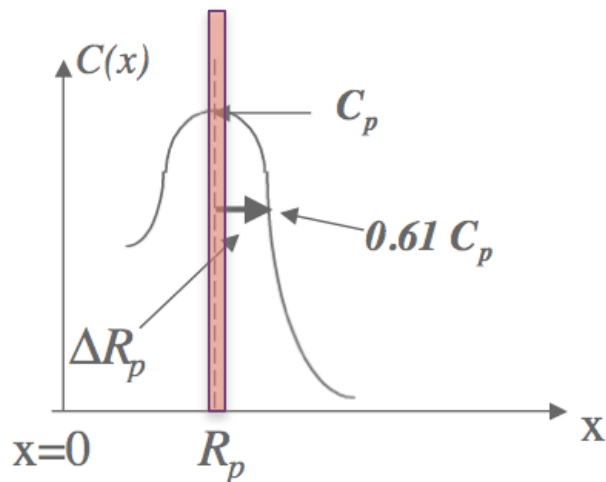


A n-type dopant atom in Silicon (P in this case) to be electrically active (i.e. to supply one electron for electrical conduction) must occupy a **substitutional site** in the perfect Si lattice. Soon after ion implantation most of P atoms are in **interstitial** positions, while the Si lattice may be heavily damaged (which leads to carrier trapping, etc.)

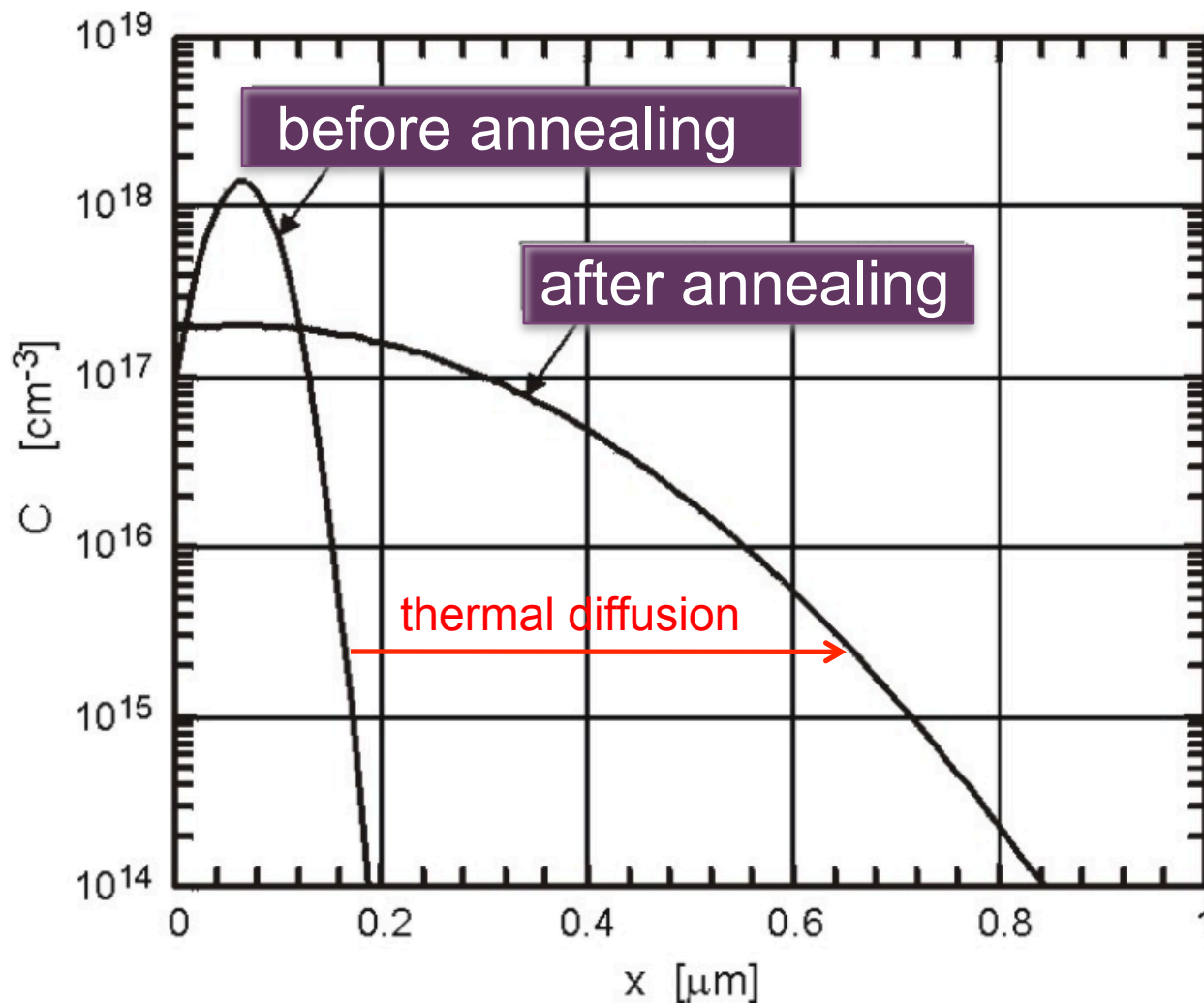
It is during subsequent high temperature **annealing** that the Si lattice recovers its crystallinity and that P atoms can rearrange to substitutional sites, thus becoming electrically active as *donors*.

Ion implantation: **CONS**

- **Straggling:** ion-target scattering at the atomic level is a *stochastic process* $\Rightarrow$ ions with same macroscopic parameters (ion species, energy, angle of incidence) penetrating a solid do not stop at the same depth, but generate a *distribution*, with a certain *spread* (the ΔR_p introduced before). This can be a limitation for those applications where an accuracy in the position of the implanted atoms smaller than the ion straggling is needed. A similar situation occurs more often today, due to the ever decreasing size of electronic devices or nanostructures.



With ion implantation it is not possible to have a thin, sharp, box-shaped distribution of dopants, like the one depicted in the example on the left. To get such a profile different techniques must be used, such as **Molecular Beam Epitaxy** (MBE).



... which causes further penetration and widening of the profile.

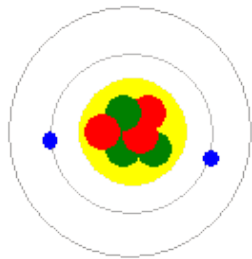
This means that most advantages of ion implantation can be lost ! To reach the best compromise, the process (implantation + damage annealing) must be carefully optimized.

Ion implantation is the technique used for most standard semiconductor doping processes in the microelectronics industry.

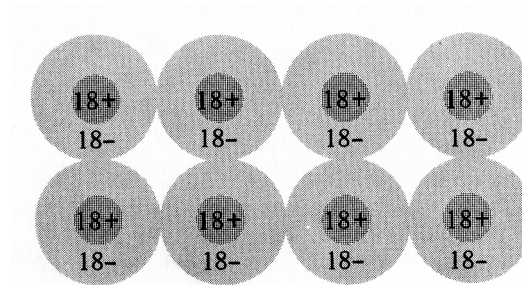
A major part of the research in this field has been on reducing implantation damage, thus maximizing the electrical activity of dopants, while minimizing diffusion, and consequent in-depth and lateral shift of electrical junctions.

This task has become more and more difficult with shrinking the size of electronic devices, today of the order of a few tens of nm.

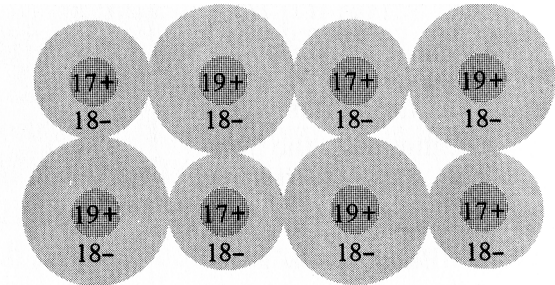
Elements of ion-solid interaction



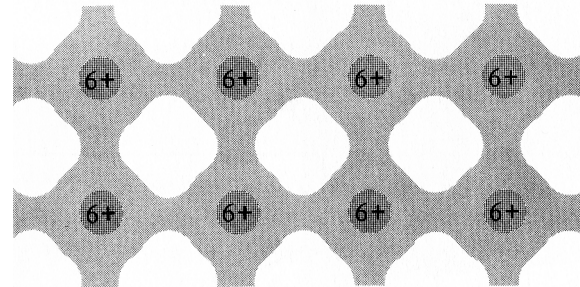
Li^+



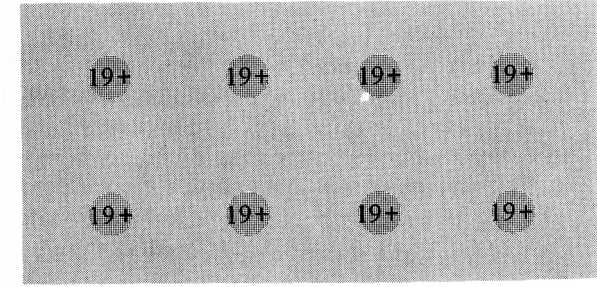
Molecular
(a)



Ionic
(b)



Covalent
(c)



Metallic
(d)

ion



solid

(i.e. an atom which has lost or gained some electrons)

(i.e. not just an ensemble of isolated atoms)